

DATA AUGMENTATION AND DYNAMIC LINEAR MODELS

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Abstract. We define a subclass of dynamic linear models with unknown hyperparameter called d -inverse-gamma models. We then approximate the marginal probability density functions of the hyperparameter and the state vector by the data augmentation algorithm of Tanner and Wong. We prove that the regularity conditions for convergence hold. For practical implementation a forward-filtering-backward-sampling algorithm is suggested, and the relation to Gibbs sampling is discussed in detail.

Keywords. Approximate Bayesian analysis; data augmentation; dynamic linear models; Gibbs sampling; Kalman filtering; state space models.

1. INTRODUCTION

In the present paper we illustrate how data augmentation methods (Tanner and Wong, 1987) arise as a natural tool of analysing dynamic linear models with unknown hyperparameter, if they are treated as incomplete data problems. We make use of the substitution algorithm suggested by Tanner and Wong (1987) to approximate the marginal probability density functions (p.d.f.s) of the hyperparameter and the state vectors.

We prove convergence of the substitution algorithm for a subclass of dynamic linear models with unknown hyperparameter which covers most dynamic linear models with unknown variances. This subclass is characterized in Section 2 by the structure of the posterior p.d.f. of the distribution of the hyperparameter conditional on the state and observation process. As this density factorizes into the product of d p.d.f.s of an inverted gamma distribution this class will be called d -inverse-gamma.

For practical implementation a Monte Carlo approach is suggested in Section 3 which is based on filtering forward and sampling trajectories of the state process backward in time ('forward-filtering-backward-sampling'). In Section 4 we discuss how this method is related to Gibbs sampling which has recently been applied to state-space models with unknown variances (Carlin *et al.*, 1992).

In Section 5 we re-analyse the UK coal consumption series (Harvey, 1989) by a basic structural model and illustrate how to assess uncertainty about the unknown variances of the model by a fully Bayesian analysis.

2. THE d -INVERSE-GAMMA CLASS

2.1. Model definition

Assume that an observed time series $y_1, \dots, y_N$ is the realization of a dynamic linear model depending on an unknown hyperparameter $\boldsymbol{\theta} \in \boldsymbol{\Theta} \subseteq \mathbb{R}^d$:

$$\mathbf{x}_t | \mathbf{x}_{t-1}, \boldsymbol{\theta} \sim N\{\mathbf{F}_t \mathbf{x}_{t-1}, \mathbf{Q}_t(\boldsymbol{\theta})\} \quad y_t | \mathbf{x}_t, \boldsymbol{\theta} \sim N\{\mathbf{H}_t \mathbf{x}_t, R_t(\boldsymbol{\theta})\} \quad t = 1, 2, \dots \quad (1)$$

$\mathbf{x}_t \in \mathbb{R}^r$ is the state vector. $R_t(\boldsymbol{\theta})$ is called the observation variance and $\mathbf{Q}_t(\boldsymbol{\theta})$ is called the process variance. The state vector $\mathbf{x}_0$ is assumed to be normally distributed: $\mathbf{x}_0 | y^0 \sim N(\hat{\mathbf{x}}_{0|0}, \mathbf{P}_{0|0})$. $\boldsymbol{\theta} = (\theta_1, \dots, \theta_d)$ is assumed to have *a priori* independent, inverted gamma-distributed components:

$$\theta_j | y^0 \sim \text{IG}(\alpha_0^{(j)}, \beta_0^{(j)}) \quad \beta_0^{(j)} > 0, j = 1, \dots, d. \quad (2)$$

$\mathbf{x}_0 | y^0$ and $\boldsymbol{\theta} | y^0$ are assumed to be independent *a priori*. Let $\mathbf{x}^N = (\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N)$ denote the whole process of state vectors from 0 to N (state process) and let $y^N = (y_1, \dots, y_N)$ denote all observations up to N .

It is well known that the conditional posterior p.d.f. $p(\mathbf{x}^N | \boldsymbol{\theta}, y^N)$ of the whole state process $\mathbf{x}^N$ given the observations y^N and the hyperparameter $\boldsymbol{\theta}$ is the density of a normal distribution (e.g. Harvey, 1989). The conditional density $p(\boldsymbol{\theta} | \mathbf{x}^N, y^N)$ of $\boldsymbol{\theta}$ given $\mathbf{x}^N$ and given all observations y^N has a known analytical form (e.g. Broemeling, 1985):

$$\begin{aligned} p(\boldsymbol{\theta} | \mathbf{x}^N, y^N) &\propto p(y_N | \mathbf{x}^N, \boldsymbol{\theta}, y^0) p(\mathbf{x}^N | \boldsymbol{\theta}, y^0) p(\boldsymbol{\theta} | y^0) \\ &= \prod_{t=1}^N p(y_t | \mathbf{x}_t, \boldsymbol{\theta}) \prod_{t=1}^N p(\mathbf{x}_t | \mathbf{x}_{t-1}, \boldsymbol{\theta}) p(\mathbf{x}_0 | y^0) p(\boldsymbol{\theta} | y^0). \end{aligned} \quad (3)$$

If $\mathbf{Q}_t(\boldsymbol{\theta})$ is regular, then (3) takes the following form:

$$\begin{aligned} p(\boldsymbol{\theta} | \mathbf{x}^N, y^N) &\propto \\ &p(\boldsymbol{\theta} | y^0) \prod_{t=1}^N |\mathbf{Q}_t(\boldsymbol{\theta})|^{-1/2} \exp \left\{ -\frac{1}{2} \sum_{t=1}^N (\mathbf{x}_t - \mathbf{F}_t \mathbf{x}_{t-1})^T \mathbf{Q}_t(\boldsymbol{\theta})^{-1} (\mathbf{x}_t - \mathbf{F}_t \mathbf{x}_{t-1}) \right\} \\ &\times \prod_{t=1}^N |R_t(\boldsymbol{\theta})|^{-1/2} \exp \left\{ -\frac{1}{2} \sum_{t=1}^N (y_t - \mathbf{H}_t \mathbf{x}_t)^2 R_t(\boldsymbol{\theta})^{-1} \right\}. \end{aligned} \quad (4)$$

Whether $p(\boldsymbol{\theta} | \mathbf{x}^N, y^N)$ is the density of a standard distribution or not depends on how the variances $\mathbf{Q}_t(\boldsymbol{\theta})$ and $R_t(\boldsymbol{\theta})$ depend on $\boldsymbol{\theta}$. We now characterize a subclass of dynamic linear models by the structure of this density.

DEFINITION 1. A path $\mathbf{x}^N$ of the state process is called *realizable* from model (1) if there exists a hyperparameter $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ such that

$$p(\mathbf{x}_1, \dots, \mathbf{x}_N | \boldsymbol{\theta}, \mathbf{x}_0) = \prod_{t=1}^N p(\mathbf{x}_t | \mathbf{x}_{t-1}, \boldsymbol{\theta}) > 0. \quad (5)$$

DEFINITION 2. A dynamic linear model with unknown hyperparameter $\theta = (\theta_1, \dots, \theta_d) \in \Theta = \mathbb{R}_+^d$ is called d -inverse-gamma if the conditional posterior p.d.f. $p(\theta | \mathbf{x}^N, \mathbf{y}^N)$ of θ given a realizable path of the state process $\mathbf{x}^N$ and given the observation process $\mathbf{y}^N$ factorizes into the product of d densities of inverted gamma distributions:

$$p(\theta | \mathbf{x}^N, \mathbf{y}^N) = \prod_{j=1}^d p(\theta_j | \mathbf{x}^N, \mathbf{y}^N) \quad \theta_j | \mathbf{x}^N, \mathbf{y}^N \sim \text{IG}\{\alpha_N^{(j)}, \beta_N^{(j)}(\mathbf{x}^N, \mathbf{y}^N)\}. \quad (6)$$

2.2. Discussion and conditional posterior analysis

The main property of this model class is the fact that the components of the hyperparameter are conditionally independent given the state and the observation process. A variety of dynamic linear models with unknown hyperparameter, e.g. the local level model, the dynamic trend model or regression models with dynamic regression coefficients, belong to the d -inverse-gamma class. In this subsection we are going to discuss structures of $Q_t(\theta)$ and $R_t(\theta)$ which ensure that the model is inverse-gamma.

Typically, $R_t(\theta)$ is of the form

$$R_t(\theta) = v_t(y^{t-1})\theta_d \quad (7)$$

with θ_d unknown and $v_t(\cdot)$ equal to a positive function which is deterministic given y^{t-1} .

As far as $Q_t(\theta)$ is concerned we shall distinguish between the ‘fully dynamic’ case of regular $Q_t(\theta)$ and the ‘partly dynamic’ case of non-regular $Q_t(\theta)$. For a fully dynamic model $Q_t(\theta)$ typically has one of the following forms:

$$Q_t(\theta) = \theta_1 A_t(y^{t-1}) \quad (8)$$

$$Q_t(\theta) = B_t(y^{t-1}) \text{diag}(\theta_1, \dots, \theta_r) B_t^T(y^{t-1}) \quad (9)$$

with $A_t(y^{t-1})$ and $B_t(y^{t-1})$ being deterministic regular matrices, if y^{t-1} is known. $\text{Diag}(\cdot)$ is a diagonal matrix. In simple cases the function $v_t(\cdot)$ in (7) and the matrices $A_t(\cdot)$ in (8) or $B_t(\cdot)$ in (9) will be constant (e.g. local level model, dynamic trend model, regression models with a dynamic regression coefficient). In some cases they are equal to a constant depending on time. More generally, they may also depend on past paths of the observation process (e.g. the conditionally Gaussian model studied in Liptser and Shirayev (1977)).

It is straightforward to determine the parameters of the conditional posterior p.d.f. $p(\theta | \mathbf{x}^N, \mathbf{y}^N)$ in (6) from (4). They are given by

$$\alpha_N^{(j)} = \alpha_0^{(j)} + \frac{N}{2}$$

$$\beta_N^{(j)}(\mathbf{x}^N, \mathbf{y}^N) = \beta_0^{(j)} + b_N^{(j)}(\mathbf{x}^N, \mathbf{y}^N) \quad j = 1, \dots, d.$$

The parameter $b_N^{(d)}(\cdot)$ which determines the posterior of the scaling factor θ_d of the observation variance is given by

$$b_N^{(d)}(\mathbf{x}^N, y^N) = \frac{1}{2} \sum_{t=1}^N \frac{(y_t - \mathbf{H}_t \mathbf{x}_t)^2}{v_t(y^{t-1})}. \quad (10)$$

The parameters determining the posterior of the other components depend on the form of $\mathbf{Q}_t(\boldsymbol{\theta})$. If $\mathbf{Q}_t(\boldsymbol{\theta})$ is of the form (8) then the model is 2-inverse-gamma with $b_N^{(1)}(\cdot)$ given by

$$b_N^{(1)}(\mathbf{x}^N, y^N) = \frac{1}{2} \sum_{t=1}^N (\mathbf{x}_t - \mathbf{F}_t \mathbf{x}_{t-1})^\top \mathbf{A}_t^{-1}(y^{t-1}) (\mathbf{x}_t - \mathbf{F}_t \mathbf{x}_{t-1}). \quad (11)$$

If $\mathbf{Q}_t(\boldsymbol{\theta})$ is of the form (9) then the model is $(r+1)$ -inverse-gamma with $b_N^{(j)}(\cdot)$, $j = 1, \dots, r$, given by

$$b_N^{(j)}(\mathbf{x}^N, y^N) = \frac{1}{2} \sum_{t=1}^N [\mathbf{B}_t^{-1}(y^{t-1}) (\mathbf{x}_t - \mathbf{F}_t \mathbf{x}_{t-1})]_j^2. \quad (12)$$

$[\cdot]_j$ denotes the j th component of a vector.

Now let us proceed with the partly dynamic model. A partly dynamic model most often occurs if in a fully dynamic model with unknown hyperparameter $\tilde{\boldsymbol{\theta}}$ m components of $\tilde{\boldsymbol{\theta}}$ are restricted to 0. Then only $r-m$ components of $\tilde{\boldsymbol{\theta}}$ – denoted by $\boldsymbol{\theta}$ – remain unknown.

LEMMA 1. *Assume that a fully dynamic model belongs to the d -inverse-gamma class with $\mathbf{Q}_t(\tilde{\boldsymbol{\theta}})$ taking the form (8) or (9). Then the partly dynamic model obtained by restricting m components of $\tilde{\boldsymbol{\theta}}$ to 0 is a $(d-m)$ -inverse-gamma model. If $\mathbf{x}^N$ is a realizable path from the restricted model, then the following holds: the parameters $b_N^{(j)}(\mathbf{x}^N, y^N)$, $j = 1, \dots, d-m$ of the conditional posterior $p(\boldsymbol{\theta} | \mathbf{x}^N, y^N)$ in the restricted model are related to the parameters $\tilde{b}_N^{(i)}(\mathbf{x}^N, y^N)$ in the unrestricted model by the following identity:*

$$b_N^{(j)}(\mathbf{x}^N, y^N) = \tilde{b}_N^{(i_j)}(\mathbf{x}^N, y^N) \quad (13)$$

with i_j such that $\theta_j = \tilde{\theta}_{i_j}$.

A proof of this lemma is given in the Appendix.

2.3. Examples

EXAMPLE 1: LOCAL LEVEL MODEL. The local level model with the one-dimensional state variable x_t equal to the level of the process (e.g. Harvey, 1989) is characterized by

$$p(x_t | x_{t-1}, \boldsymbol{\theta}) \sim N(x_{t-1}, \theta_1) \quad p(y_t | x_t, \boldsymbol{\theta}) \sim N(x_t, \theta_2) \quad t = 1, 2, \dots$$

This model is 2-inverse-gamma with $v_t(y^{t-1}) = 1$ and $A_t(y^{t-1}) = 1$ in (8), if $\boldsymbol{\theta} = (\theta_1, \theta_2)$ is unknown. Given an inverted gamma prior for $\theta_1 | y^0$ and $\theta_2 | y^0$

and given a normal prior for $x_0|y^0$, the conditional posterior $p(\theta|x^N, y^N)$ is the product of two independent inverted gamma densities with the parameters in (10) and (11) given by

$$\alpha_N^{(j)} = \alpha_0^{(j)} + \frac{N}{2} \quad j = 1, 2$$

$$\beta_N^{(1)} = \beta_0^{(1)} + \frac{1}{2} \sum_{t=1}^N (x_t - x_{t-1})^2 \quad \beta_N^{(2)} = \beta_0^{(2)} + \frac{1}{2} \sum_{t=1}^N (y_t - x_t)^2.$$

EXAMPLE 2: DYNAMIC TREND MODEL. The dynamic trend model (see, for example, Harrison and Stevens, 1976) has a two-dimensional state vector $x_t = (\mu_t, a_t)$. μ_t is the mean of the observation density and a_t is the trend component. The model is characterized by (1) with

$$F_t = F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad Q_t(\theta) = F \text{diag}(\theta_1, \theta_2) F^T$$

$$H_t = (1 \ 0) \quad R_t(\theta) = \theta_3.$$

This model is 3-inverse-gamma with $v_t(y^{t-1}) = 1$ and $B_t(y^{t-1}) = F$ in (9), if $\theta = (\theta_1, \theta_2, \theta_3)$ is unknown. Given an inverted gamma prior for each component of θ and given a normal prior for $x_0|y^0$, the conditional posterior $p(\theta|x^N, y^N)$ is the product of three inverted gamma densities with the parameters in (10) and (11) given by

$$\alpha_N^{(j)} = \alpha_0^{(j)} + \frac{N}{2} \quad j = 1, 2, 3 \quad \beta_N^{(1)} = \beta_0^{(1)} + \frac{1}{2} \sum_{t=1}^N \{\mu_{t-1} - (\mu_{t-1} + a_t)\}^2$$

$$\beta_N^{(2)} = \beta_0^{(2)} + \frac{1}{2} \sum_{t=1}^N (a_t - a_{t-1})^2 \quad \beta_N^{(3)} = \beta_0^{(3)} + \frac{1}{2} \sum_{t=1}^N (y_t - \mu_t)^2.$$

From this model we obtain a partly dynamic model if the dynamic trend component is restricted to a static one: $\tilde{\theta}_2 = 0$ – note the change of notation from θ to $\tilde{\theta}$. The new model is 2-inverse-gamma with $\theta_1 = \tilde{\theta}_1$ and $\theta_2 = \tilde{\theta}_3$. The second component of w_t is 0 iff $a_t = a \ \forall t = 0, \dots, N$. Thus a path x^N is realizable iff the second component of x_t is the same $\forall t = 0, \dots, N$. Therefore $\beta_N^{(1)}$ is given by

$$\beta_N^{(1)} = \beta_0^{(1)} + \frac{1}{2} \sum_{t=1}^N \{\mu_t - (\mu_{t-1} + a)\}^2$$

and $\beta_N^{(2)} = \tilde{\beta}_N^{(3)}$.

EXAMPLE 3: BASIC STRUCTURAL MODEL. The basic structural model defined in Harvey (1989) may be written as a dynamic linear model with a $(k+1)$ -dimensional state vector $x_t = (\mu_t, a_t, s_{t-k+1}, \dots, s_t)$ where k denotes the number of seasons, μ_t the trend component, s_t the seasonal component and a_t the slope of the local trend function at time t . The model is defined by

the following transition densities:

$$\begin{aligned}\mu_t | \mathbf{x}_{t-1} &\sim N(\mu_{t-1} + a_{t-1}, \theta_1) \\ a_t | \mathbf{x}_{t-1} &\sim N(a_{t-1}, \theta_2) \\ s_t | \mathbf{x}_{t-1} &\sim N\left(-\sum_{j=1}^{k-1} s_{t-j}, \theta_3\right) \\ y_t | \mathbf{x}_t &\sim N(\mu_t + s_t, \theta_4).\end{aligned}$$

Since for the remaining components $x_{t,j}$, $3 \leq j \leq k$, the transition densities are Dirac densities, this model is not a ‘fully dynamic’ one with $\mathbf{Q}_t(\boldsymbol{\theta})$ regular. It may, however, be viewed as a partly dynamic model, obtained by restricting the $k-2$ variances $\theta_5, \dots, \theta_{k+2}$ in the fully dynamic model

$$x_{t,j} | \mathbf{x}_{t-1} \sim N(x_{t-1,j}, \theta_{2+j}) \quad 3 \leq j \leq k$$

to 0. Therefore from Lemma 1 we deduce that the basic structural model is a 4-inverse-gamma model with the parameters (10) and (11) of the conditional posterior given by

$$\begin{aligned}\alpha_N^{(j)} &= \alpha_0^{(j)} \quad 1 \leq j \leq 4 \\ \beta_N^{(1)} &= \beta_0^{(1)} + \frac{1}{2} \sum_{t=1}^N \{\mu_t - (\mu_{t-1} + a_{t-1})\}^2 \\ \beta_N^{(2)} &= \beta_0^{(2)} + \frac{1}{2} \sum_{t=1}^N (a_t - a_{t-1})^2 \\ \beta_N^{(3)} &= \beta_0^{(3)} + \frac{1}{2} \sum_{t=1}^N \left(\sum_{j=0}^{k-1} s_{t-j} \right)^2 \\ \beta_N^{(4)} &= \beta_0^{(4)} + \frac{1}{2} \sum_{t=1}^N \{y_t - (\mu_t + s_t)\}^2.\end{aligned}$$

More specific models may be derived from the basic structural model by restricting some variances to 0. Lemma 1 then guarantees that all of them are inverse-gamma models.

3. APPROXIMATE POSTERIOR DENSITIES

We now turn to the problem of approximating the marginal density $p(\boldsymbol{\theta} | y^N)$ of the unknown hyperparameter $\boldsymbol{\theta}$ and the smoothing densities $p(\mathbf{x}_t | y^N)$ which are marginal densities, too. It should be noted that for $d = 1$ the model belongs to the well-known normal-inverse-gamma class. This is typically the case when the dynamic model degenerates to a static one with $\mathbf{x}_t \equiv \mathbf{x}$. Then both the posterior $p(\boldsymbol{\theta} | y^N)$ of the unknown hyperparameter $\boldsymbol{\theta}$ and the

posterior density $p(\mathbf{x}|y^N)$ are known analytically (see, for example, Tanner, 1991):

$$\theta|y^N \sim \text{IG}\left\{\alpha_0^{(1)} + \frac{N-r}{2}, \beta_N^{(1)}(\hat{\mathbf{x}}^N)\right\}$$

with $\hat{\mathbf{x}}^N$ being the posterior mean of $p(\mathbf{x}|y^N, \theta = 1)$ with uninformative prior $\hat{\mathbf{x}}_{0|0} = 0$ and $\mathbf{P}_{0|0}^{-1} = 0$.

For $d > 1$ all these densities have no tractable analytical form and need to be approximated. Dynamic linear models with unknown hyperparameter are discussed, for example, in Harrison and Stevens (1976) and Harvey (1989). A number of authors have approached the problem of unknown hyperparameter from a frequentistic point of view (e.g. Harvey, 1989) by maximizing the likelihood of the hyperparameter either numerically or by the EM algorithm. The Bayesian approach dates back to the early paper of Magill (1965) and was discussed in detail by Harrison and Stevens (1976). In both papers the posterior of the hyperparameter is approximated by a discretization method called multi-process filtering. This method has been refined by Pole (1988). One of the main problems with the approach 'is the reconstruction of the continuous posterior density from the discrete approximation' (see also West and Harrison, 1989, p. 533).

In this paper we treat dynamic linear models with unknown hyperparameter as a special case of what has been called an incomplete data problem (Dempster *et al.*, 1977). We then apply the data augmentation algorithm proposed by Tanner and Wong (1987) for approximating marginal densities in incomplete data problems to our problem. This algorithm is reviewed in Section 3.1. In Section 3.2 we suggest a 'forward-filtering-backward-sampling' algorithm for practical implementation. In Section 4 we discuss how this algorithm is related to Gibbs sampling which has recently been applied to state space models (Carlin *et al.*, 1992).

3.1. The data augmentation algorithm

Tanner and Wong (1987) observed that in incomplete data problems the marginal distribution is the fixed point solution of an integral equation. If this result is applied to our problem we obtain that the posterior $p(\theta|y^N)$ of the unknown hyperparameter θ is a fixed point solution of the following integral equation:

$$p(\theta|y^N) = \iint p(\theta|\mathbf{x}^N, y^N)p(\mathbf{x}^N|\theta', y^N)p(\theta'|y^N) d\mathbf{x}^N d\theta'. \quad (14)$$

Tanner and Wong (1987) used the substitution algorithm in order to find the solution of this integral equation. Given some approximation $g_{n-1}(\theta)$ of the posterior $p(\theta|y^N)$ one may use Equation (14) to improve it:

$$g_n(\theta) = \int K(\theta, \theta')g_{n-1}(\theta') d\theta' \quad (15)$$

$$K(\boldsymbol{\theta}, \boldsymbol{\theta}') = \int p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N) p(\mathbf{x}^N|\boldsymbol{\theta}', \mathbf{y}^N) d\mathbf{x}^N. \quad (16)$$

Iteration (15) converges in L_1 -norm to a unique solution which is equal to the marginal posterior $p(\boldsymbol{\theta}|\mathbf{y}^N)$ under the following regularity conditions on $K(\boldsymbol{\theta}, \boldsymbol{\theta}')$ (Tanner and Wong, 1987, p. 538): $K(\boldsymbol{\theta}, \boldsymbol{\theta}')$ is uniformly bounded (C1); $K(\boldsymbol{\theta}, \boldsymbol{\theta}')$ is equicontinuous in $\boldsymbol{\theta}$ (C2); $\forall \boldsymbol{\theta}_0 \in \boldsymbol{\Theta}$: $\exists$ open neighbourhood $U_\varepsilon(\boldsymbol{\theta}_0)$: $K(\boldsymbol{\theta}, \boldsymbol{\theta}') > 0$, $\forall \boldsymbol{\theta}, \boldsymbol{\theta}' \in U_\varepsilon(\boldsymbol{\theta}_0)$ (C3).

To prove convergence of the substitution algorithm to the posterior of the hyperparameter for the d -inverse-gamma class we only need the following condition on the prior parameters of $\boldsymbol{\theta}$: $\beta_0^{(j)} > 0$, $j = 1, \dots, d$ (C4).

LEMMA 2. *Under Condition (C4) L_1 -convergence of $g_n(\boldsymbol{\theta})$ to the posterior $p(\boldsymbol{\theta}|\mathbf{y}^N)$ of the hyperparameter holds for d -inverse-gamma dynamic linear models.*

A proof of this lemma is given in the Appendix.

3.2. Practical implementation by forward-filtering-backward-sampling

Integration in (15) cannot be performed analytically for dynamic linear models. For such cases Tanner and Wong (1987) suggested using Monte Carlo methods for practical implementation of the substitution algorithm. In this section we discuss how such a Monte Carlo approach may be implemented for dynamic linear models. Monte Carlo integration of (15) is carried out by importance sampling from $w(\boldsymbol{\theta}, \mathbf{x}^N) = p(\mathbf{x}^N|\boldsymbol{\theta}, \mathbf{y}^N)g_{n-1}(\boldsymbol{\theta})$. Based on a sample $\boldsymbol{\theta}^{(1)}, (\mathbf{x}^N)^{(1)}, \dots, \boldsymbol{\theta}^{(M_n)}, (\mathbf{x}^N)^{(M_n)}$ of size M_n , $g_n(\boldsymbol{\theta})$ is estimated by

$$g_n(\boldsymbol{\theta}) \approx \frac{1}{M_n} \sum_{m=1}^{M_n} p\{\boldsymbol{\theta} | (\mathbf{x}^N)^{(m)}, \mathbf{y}^N\}. \quad (17)$$

Note that $g_n(\boldsymbol{\theta})$ is approximated analytically by a density from a finite mixture distribution. The type of the mixture depends on the form of the conditional posterior $p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N)$. For d -inverse-gamma dynamic linear models this density is a mixture of products of d inverted gamma densities. We determine the moments of the conditional p.d.f. $p\{\boldsymbol{\theta} | (\mathbf{x}^N)^{(m)}, \mathbf{y}^N\}$ from (6). Once these moments are calculated the simulated path of the process need not be stored.

Sampling from $w(\boldsymbol{\theta}, \mathbf{x}^N) = p(\mathbf{x}^N|\boldsymbol{\theta}, \mathbf{y}^N)g_{n-1}(\boldsymbol{\theta})$ is done by the method of composition (e.g. Tanner, 1991): first we sample a hyperparameter $\boldsymbol{\theta}^{(m)}$ from $g_{n-1}(\boldsymbol{\theta})$ and then we sample a path $(\mathbf{x}^N)^{(m)}$ of the state process from the conditional smoothing density $p(\mathbf{x}^N|\boldsymbol{\theta}^{(m)}, \mathbf{y}^N)$.

Sampling of $\boldsymbol{\theta}^{(m)}$ from $g_{n-1}(\boldsymbol{\theta})$ is easy. We just sample the number I_m of one of the members in the mixture (17) which is uniformly distributed between 1 and M_{n-1} and then sample $\boldsymbol{\theta}^{(m)}$ from the corresponding conditional density $p\{\boldsymbol{\theta} | (\mathbf{x}^N)^{(I_m)}, \mathbf{y}^N\}$. For d -inverse-gamma dynamic linear models this

density factorizes into d inverted gamma densities. Thus the components $\theta_i^{(m)}$, $i = 1, \dots, d$, of $\boldsymbol{\theta}^{(m)}$ are sampled independently.

The more difficult point is sampling a realizable path $(\mathbf{x}^N)^{(m)}$ from the smoothing density $p(\mathbf{x}^N | \mathbf{y}^N, \boldsymbol{\theta})$ which for $\boldsymbol{\theta}$ known is a normal density. We consider the following conditional representation of a multivariate normal density:

$$p(\mathbf{x}^N | \mathbf{y}^N, \boldsymbol{\theta}) = p(\mathbf{x}_N | \mathbf{y}^N, \boldsymbol{\theta}) \prod_{t=0}^{N-1} p(\mathbf{x}_t | \mathbf{x}_{t+1}, \dots, \mathbf{x}_N, \mathbf{y}^N, \boldsymbol{\theta}). \quad (18)$$

Using Bayes' theorem Equation (18) is expressed as

$$p(\mathbf{x}^N | \mathbf{y}^N, \boldsymbol{\theta}) \propto p(\mathbf{x}_N | \mathbf{y}^N, \boldsymbol{\theta}) \prod_{t=0}^{N-1} p(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{y}^t, \boldsymbol{\theta}) \quad (19)$$

$$p(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{y}^t, \boldsymbol{\theta}) \propto p(\mathbf{x}_{t+1} | \boldsymbol{\theta}, \mathbf{x}_t) p(\mathbf{x}_t | \mathbf{y}^t, \boldsymbol{\theta}). \quad (20)$$

Equations (19) and (20) motivate what will be called a forward-filtering-backward-sampling algorithm:

(a) We determine the moments $\hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})$ and $\mathbf{P}_{t|t}(\boldsymbol{\theta})$ of the filtering densities $p(\mathbf{x}_t | \mathbf{y}^t, \boldsymbol{\theta})$ by a Kalman filter running from $t = 1$ to $t = N$.

(b) We then start sampling with the latest value $\mathbf{x}_N^{(m)}$ of the state vector from the marginal $p(\mathbf{x}_N | \mathbf{y}^N, \boldsymbol{\theta}) \sim N\{\hat{\mathbf{x}}_{N|N}(\boldsymbol{\theta}), \mathbf{P}_{N|N}(\boldsymbol{\theta})\}$.

(c) We then sample $\mathbf{x}_t^{(m)}$ from the density $p(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{y}^t, \boldsymbol{\theta})$ back in time for $t = N - 1$ to $t = 0$.

There exist various ways to implement the third step. For a fully dynamic model we sample directly from the conditional density, the moments of which are given in Proposition 1.

PROPOSITION 1. Assume $\mathbf{Q}_{t+1}(\boldsymbol{\theta})$ is regular. Then the conditional densities $p(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{y}^t, \boldsymbol{\theta})$, $t = 0, \dots, N - 1$, in (20) are normal densities with moments $\hat{\mathbf{x}}_{t|N}(\mathbf{x}_{t+1}, \boldsymbol{\theta})$ and $\mathbf{P}_{t|N}(\boldsymbol{\theta})$ given by

$$\hat{\mathbf{x}}_{t|N}(\mathbf{x}_{t+1}, \boldsymbol{\theta}) = \{\mathbf{I} - \mathbf{A}_{t+1}(\boldsymbol{\theta})\mathbf{F}_{t+1}\}\hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta}) + \mathbf{A}_{t+1}(\boldsymbol{\theta})\mathbf{x}_{t+1}$$

$$\mathbf{P}_{t|N}(\boldsymbol{\theta}) = \{\mathbf{I} - \mathbf{A}_{t+1}(\boldsymbol{\theta})\mathbf{F}_{t+1}\}\mathbf{P}_{t|t}(\boldsymbol{\theta})$$

$$\mathbf{A}_{t+1}(\boldsymbol{\theta}) = \mathbf{P}_{t|t}(\boldsymbol{\theta})\mathbf{F}_{t+1}^\top \{\mathbf{F}_{t+1}\mathbf{P}_{t|t}(\boldsymbol{\theta})\mathbf{F}_{t+1}^\top + \mathbf{Q}_{t+1}(\boldsymbol{\theta})\}^{-1}$$

where $\hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})$ and $\mathbf{P}_{t|t}(\boldsymbol{\theta})$ are the moments of the filtering densities $p(\mathbf{x}_t | \boldsymbol{\theta}, \mathbf{y}^t)$ obtained by Kalman filtering.

Note that computing the conditional variance by Proposition 1 involves the inversion of an r -dimensional matrix at each time step. For a partly dynamic model with $d = \text{rank}\{\mathbf{Q}_{t+1}(\boldsymbol{\theta})\} < r$ the conditional density $p(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{y}^t, \boldsymbol{\theta})$ is degenerate. Although Proposition 1 will work nevertheless, a more efficient algorithm is derived in Proposition 2, which involves the inversion of a d -dimensional matrix only. The basic idea is to transform the state vector $\mathbf{x}_t$ to a new variable $\mathbf{z}_t$ which has only d random components; the remaining

components of $\mathbf{z}_t$ are deterministic given $\mathbf{x}_{t+1}$. We then sample the random components of $\mathbf{z}_t$ and transform back to $\mathbf{x}_t$. For some models of practical importance this leads to a substantial reduction in computing time. Consider, for example, a basic structural model (Example 3) with the number of seasons equal to 12, where instead of a 13-dimensional matrix we have to invert only a three-dimensional matrix.

PROPOSITION 2. Assume $\mathbf{F}_{t+1}(\boldsymbol{\theta})$ is regular whereas $\mathbf{Q}_{t+1}(\boldsymbol{\theta})$ is singular. Let $\boldsymbol{\Pi} \in \mathbb{R}^{r \times r}$ be a permutation matrix such that

$$\boldsymbol{\Pi} \mathbf{Q}_{t+1}(\boldsymbol{\theta}) \boldsymbol{\Pi}^T = \begin{pmatrix} \mathbf{Q}_{t+1}^*(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \quad (21)$$

where $\mathbf{Q}_{t+1}^*(\boldsymbol{\theta}) \in \mathbb{R}^{d \times d}$. Then the transformation

$$\mathbf{z}_t = \boldsymbol{\Pi} \mathbf{F}_{t+1} \mathbf{x}_t$$

yields a variable $\mathbf{z}_t$ which, conditional on $\mathbf{x}_{t+1}$ and y^t , consists of d stochastic and $r - d$ deterministic components:

$$E(\mathbf{z}_t | \mathbf{x}_{t+1}, y^t, \boldsymbol{\theta}) = \begin{pmatrix} \hat{\mathbf{z}}_{t|N}^*(\mathbf{x}_{t+1}, \boldsymbol{\theta}) \\ \boldsymbol{\Pi}^0 \mathbf{x}_{t+1} \end{pmatrix} \quad V(\mathbf{z}_t | \mathbf{x}_{t+1}, y^t, \boldsymbol{\theta}) = \begin{pmatrix} \mathbf{P}_{t|N}^*(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$$

where

$$\begin{aligned} \hat{\mathbf{z}}_{t|N}^*(\mathbf{x}_{t+1}, \boldsymbol{\theta}) &= \{\mathbf{I} - \mathbf{A}_{t+1}^*(\boldsymbol{\theta})\} \boldsymbol{\Pi}^* \mathbf{F}_{t+1} \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta}) + \mathbf{A}_{t+1}^*(\boldsymbol{\theta}) \boldsymbol{\Pi}^* \mathbf{x}_{t+1} \\ \mathbf{P}_{t|N}^*(\boldsymbol{\theta}) &= \{\mathbf{I} - \mathbf{A}_{t+1}^*(\boldsymbol{\theta})\} \boldsymbol{\Sigma}_{t+1}^*(\boldsymbol{\theta}) \\ \boldsymbol{\Sigma}_{t+1}^*(\boldsymbol{\theta}) &= \boldsymbol{\Pi}^* \mathbf{F}_{t+1} \mathbf{P}_{t|t}(\boldsymbol{\theta}) (\boldsymbol{\Pi}^* \mathbf{F}_{t+1})^T \\ \mathbf{A}_{t+1}^*(\boldsymbol{\theta}) &= \boldsymbol{\Sigma}_{t+1}^*(\boldsymbol{\theta}) \{\boldsymbol{\Sigma}_{t+1}^*(\boldsymbol{\theta}) + \mathbf{Q}_{t+1}^*(\boldsymbol{\theta})\}^{-1} \end{aligned}$$

and $\boldsymbol{\Pi}^*$ equals the first d rows and $\boldsymbol{\Pi}^0$ the last $r - d$ rows of $\boldsymbol{\Pi}$. A sample $\mathbf{x}_t^{(m)}$ from $p(\mathbf{x}_t | \mathbf{x}_{t+1}, y^t, \boldsymbol{\theta})$ is given by

$$\mathbf{x}_t^{(m)} = \mathbf{F}_{t+1}^{-1} \boldsymbol{\Pi}^T \begin{pmatrix} (\mathbf{z}_t^{(m)}) \\ \boldsymbol{\Pi}^0 \mathbf{x}_{t+1} \end{pmatrix} \quad (\mathbf{z}_t^{(m)}) \sim N\{\hat{\mathbf{z}}_{t|N}^*(\mathbf{x}_{t+1}, \boldsymbol{\theta}), \mathbf{P}_{t|N}^*(\boldsymbol{\theta})\}.$$

It should be mentioned that with both propositions numerical problems may arise for the very first time steps if we model vague prior information on the state variable by large prior variances. With vague (large variances) or improper ($\mathbf{P}_{0|0}^{-1}(\boldsymbol{\theta}) = \mathbf{0}$) priors we find the following modification useful. We determine the filtering densities $p(\mathbf{x}_t | y^t, \boldsymbol{\theta})$ from $t = 1$ to $t = K$, with K as small as possible, by an information filter (see, for example, Anderson and Moore, 1979, p. 138),

$$\begin{aligned} \mathbf{m}_{t|t}(\boldsymbol{\theta}) &= (\mathbf{F}_t^{-1})^T \mathbf{V}_t(\boldsymbol{\theta}) \mathbf{m}_{t-1|t-1}(\boldsymbol{\theta}) + \mathbf{H}_t^T \mathbf{R}_t^{-1}(\boldsymbol{\theta}) y_t \\ \mathbf{C}_{t|t}(\boldsymbol{\theta}) &= (\mathbf{F}_t^{-1})^T \mathbf{V}_t(\boldsymbol{\theta}) \mathbf{C}_{t-1|t-1}(\boldsymbol{\theta}) \mathbf{F}_t^{-1} + \mathbf{H}_t^T \mathbf{R}_t^{-1}(\boldsymbol{\theta}) \mathbf{H}_t \\ \mathbf{V}_t(\boldsymbol{\theta}) &= \{\mathbf{I} + \mathbf{C}_{t-1|t-1}(\boldsymbol{\theta}) \mathbf{W}_t(\boldsymbol{\theta})\}^{-1} = \{\mathbf{I} + \mathbf{W}_t(\boldsymbol{\theta}) \mathbf{C}_{t-1|t-1}(\boldsymbol{\theta})\}^{-1} \\ \mathbf{W}_t(\boldsymbol{\theta}) &= \mathbf{F}_t^{-1} \mathbf{Q}_t(\boldsymbol{\theta}) (\mathbf{F}_t^{-1})^T, \end{aligned} \quad (22)$$

and only for $t \geq K$ do we use the Kalman filter with $\mathbf{P}_{K|K}(\boldsymbol{\theta}) = \mathbf{C}_{K|K}^{-1}(\boldsymbol{\theta})$ and $\hat{\mathbf{x}}_{K|K}(\boldsymbol{\theta}) = \mathbf{P}_{K|K}(\boldsymbol{\theta})\mathbf{m}_{K|K}(\boldsymbol{\theta})$. We sample back from $t = N$ until $t = K$ by Proposition 1 or Proposition 2 and for $t = K - 1, \dots, 0$ we switch to Proposition 3 which expresses the moments of the conditional densities in terms of the moments of the information filter.

PROPOSITION 3. The moments of the conditional densities $p(\mathbf{x}_t | \mathbf{x}_{t+1}, y^t, \boldsymbol{\theta})$ in (20) may be expressed in terms of the moments given by the information filter

$$\begin{aligned}\mathbf{P}_{t|N}(\boldsymbol{\theta}) &= \mathbf{W}_{t+1}(\boldsymbol{\theta})\mathbf{V}_{t+1}(\boldsymbol{\theta}) \\ \hat{\mathbf{x}}_{t|N}(\mathbf{x}_{t+1}, \boldsymbol{\theta}) &= \mathbf{P}_{t|N}(\boldsymbol{\theta})\mathbf{m}_{t|t}(\boldsymbol{\theta}) + \mathbf{V}_{t+1}^T(\boldsymbol{\theta})\mathbf{F}_{t+1}^{-1}\mathbf{x}_{t+1}.\end{aligned}$$

All propositions are proved in the Appendix. To conclude this section, let us mention that an approximation of the marginal smoothing densities $p(\mathbf{x}_t | y^N)$ of the state vector is given by Monte Carlo integration:

$$p(\mathbf{x}_t | y^N) = \int p(\mathbf{x}_t | \boldsymbol{\theta}, y^N) p(\boldsymbol{\theta} | y^N) d\boldsymbol{\theta}. \quad (23)$$

It is not necessary to compute these densities during iteration for the posterior $p(\boldsymbol{\theta} | y^N)$ of the hyperparameter. Having found an appropriate approximation $g_n(\boldsymbol{\theta})$ of $p(\boldsymbol{\theta} | y^N)$ the smoothing density of the state vector $\mathbf{x}_t | y^N$ is approximated by a mixture of Gaussian densities:

$$p(\mathbf{x}_t | y^N) \approx \frac{1}{M_X} \sum_{m=1}^{M_X} p(\mathbf{x}_t | \boldsymbol{\theta}^{(m)}, y^N) \quad (24)$$

where $\boldsymbol{\theta}^{(m)}$, $m = 1, \dots, M_X$, is sampled from the approximate importance function $g_n(\boldsymbol{\theta})$ as described above.

4. DISCUSSION AND RELATION TO GIBBS SAMPLING

In a recent paper (Carlin *et al.*, 1992) Gibbs sampling has been applied to the analysis of state-space models with unknown variances. It is interesting to note that when data augmentation is implemented by the forward-filtering-backward-sampling method of Section 3.2 with $M_n = 1$ the resulting algorithm is effectively one run of Gibbs sampling: $\boldsymbol{\theta}^{(0)}$ is sampled from $g_0(\boldsymbol{\theta} | y^N)$, $(\mathbf{x}^N)^{(0)}$ from $p(\mathbf{x}^N | \boldsymbol{\theta}^{(0)}, y^N)$, $\boldsymbol{\theta}^{(1)}$ from $g_1(\boldsymbol{\theta} | y^N) = p(\boldsymbol{\theta} | (\mathbf{x}^N)^{(0)}, y^N)$, $(\mathbf{x}^N)^{(1)}$ from $p(\mathbf{x}^N | \boldsymbol{\theta}^{(1)}, y^N)$ and so on. This insight indicates how to implement Gibbs sampling in cases which are not covered by the lemma in Carlin *et al.* (1992, p. 495). As this lemma requires system variances $\mathbf{Q}_t(\boldsymbol{\theta})$ to be regular it cannot be applied to partly dynamic models such as the basic structural model (see Section 2.4). For such models Proposition 2 which is not restricted to fully dynamic models is useful even for a pure Gibbs sampling approach.

Data augmentation and Gibbs sampling are similar in so far as both deal

with the augmented posterior $p(\mathbf{x}^N, \boldsymbol{\theta} | y^N)$ and approximate the marginals $p(\boldsymbol{\theta} | y^N)$ by a density estimate which is derived by Monte Carlo integration of (15). The main difference between the two approaches lies in the way this Monte Carlo integration is carried out. In Carlin *et al.* (1992) the density estimate (17) is derived by standard importance sampling with $\boldsymbol{\theta}^{(m)}$, $(\mathbf{x}^N)^{(m)}$, $m = 1, \dots, M_G$, drawn from the complicated joint distribution $p(\mathbf{x}^N, \boldsymbol{\theta} | y^N)$ by Gibbs sampling of length L_G . With Gibbs sampling the total number of drawings of $(\mathbf{x}^N, \boldsymbol{\theta})$ is equal to $L_G M_G$.

Data augmentation implemented by the forward-filtering-backward-sampling method applies Monte Carlo integration with adaptive importance sampling. The sample $\boldsymbol{\theta}^{(m)}$, $(\mathbf{x}^N)^{(m)}$, $m = 1, \dots, M_n$, is drawn recursively from the approximate importance function $p(\mathbf{x}^N | \boldsymbol{\theta}, y^N) g_{n-1}(\boldsymbol{\theta})$ by the method of composition. The resulting density $g_n(\boldsymbol{\theta})$ determines the new importance function and so on. Lemma 2 guarantees that after a sufficient number L_D of iterations we sample from the correct density. M_n influences the accuracy of Monte Carlo integration with respect to the approximate importance function $g_{n-1}(\boldsymbol{\theta})$. As long as $g_{n-1}(\boldsymbol{\theta})$ is far from $p(\boldsymbol{\theta} | y^N)$ there is no use in making this integral accurate, and M_n may be rather small until the steady state is reached. The adaptive nature of the algorithm makes it easy to assess convergence by monitoring, for example, the quantiles of the sample drawn from $g_{n-1}(\boldsymbol{\theta})$ (see Tanner and Wong, 1987). Once the steady state is reached, M_n is increased for the last few iterations to reduce the variance of the Monte Carlo density estimate. The total number of drawings of $(\mathbf{x}^N, \boldsymbol{\theta})$ is equal to $\sum_{n=1}^{L_D} M_n$. This may be much smaller than the corresponding quantity for Gibbs sampling, as is illustrated by the following example.

Consider a time series generated by the local level model (Example 1) with $\theta_1 = 1$ and $\theta_2 = 1$. Assume that the starting values are very poor: for Gibbs sampling $\boldsymbol{\theta}^{(0)} = (0.01, 10)$ and for data augmentation $g_0(\boldsymbol{\theta}) = p_{IG}(\theta_1; 3, 0.02)p_{IG}(\theta_2; 3, 20)$, implying that $E_0(\boldsymbol{\theta} | y^N) = (0.01, 10)$. Gibbs sampling is implemented in a parallel processing mode with $M_G = 100$. Data augmentation is implemented with $M_n = 100$.

Figure 1 shows the 5%, 50% and 95% quantile of the approximate posterior as a function of the number n of iterations. The left-hand plots show the quantiles of the sample $\boldsymbol{\theta}_n^{(m)}$, $m = 1, \dots, M_G$, obtained by Gibbs sampling as a function of n . The poor starting values lead to an extremely long burn-in phase (about 750 iterations!) before the steady state is reached. The right-hand side plots show the corresponding quantiles of the samples drawn from the approximate posterior $g_n(\boldsymbol{\theta})$ obtained by data augmentation as a function of n . Despite the poor starting approximation the steady state is reached after only about 50 iterations. The total number of drawings to reach the steady state is about 5000 compared with about 75 000. This tremendous saving is due to the adaptive nature of data augmentation which makes an efficient use of all previous drawings whereas with the parallel non-adaptive mode of Gibbs sampling a lot of information is lost.

As a final remark, we should mention that it is not straightforward to

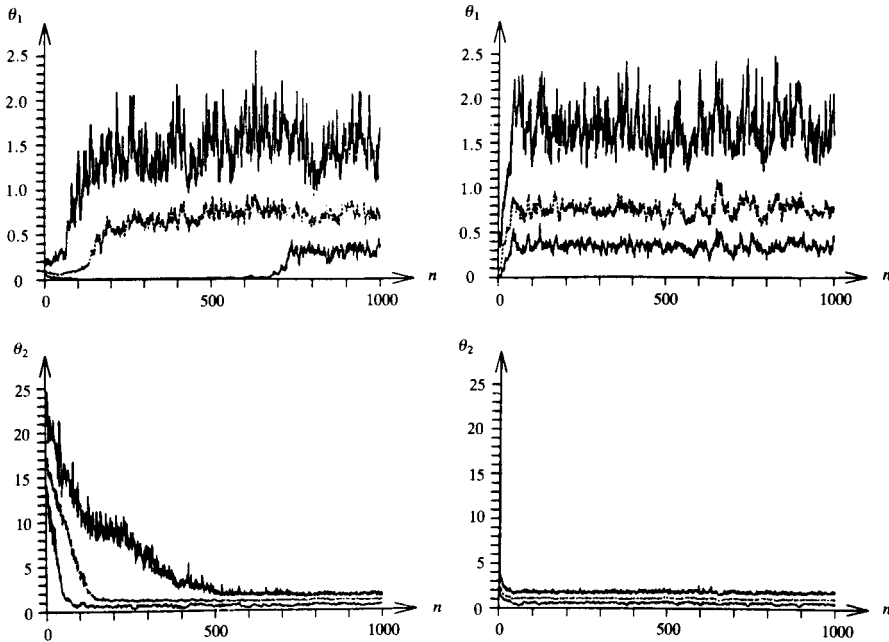


FIGURE 1. Monitoring convergence of the approximate posterior for Gibbs sampling (left) and data augmentation (right) for a local level model with $\theta_1 = 1$ and $\theta_2 = 1$.

extend the forward-filtering-backward-sampling algorithm to non-linear and non-normal state-space models. For such models Gibbs sampling turns out to be very useful. For linear and normal state-space models, however, we claim that the forward-filtering-backward-sampling algorithm is superior, because it exploits the conditionally linear structure of the model.

5. AN ILLUSTRATIVE CASE STUDY

In this section we shall re-analyse the UK coal consumption series (92 quarterly observations from 1960 to 1982) modelled in Harvey (1989, p. 95) by a quarterly basic structural model. There the unknown hyperparameter $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$ is estimated by maximum likelihood (ML) methods; the standard deviations are estimated by the inverse of the Fisher information which under certain regularity conditions is the squared standard deviation of the asymptotic normal density:

	θ_1	θ_2	θ_3	θ_4
ML estimate	0.000642	0.000029	0.000084	0.0114
Standard deviations	0.001126	0.000025	0.000058	0.0025

For the UK coal consumption series this normal approximation leads for all variances but θ_4 to non-negligible probabilities of negative variances. Such a result may occur if the likelihood and therefore the posterior are far from normality because the number of observations is still too small. It may also indicate that asymptotic normality does not hold because some variances are actually zero and the regularity conditions are violated. A fully Bayesian analysis copes with both situations. It should be emphasized that even the second non-regular situation is covered by the present paper, as convergence of our algorithm also holds for such a case provided that Condition (C4) – $\beta_0^{(j)} > 0, \forall j$ – is fulfilled.

The fully Bayesian analysis combines the likelihood with a prior on θ which has the structure given in (2). We have tried to choose the prior in such a way that on the one hand it is rather uninformative and on the other hand Condition (C4) holds:

$$\theta_j | y^0 \sim \text{IG}(0.01, 10^{-5}) \quad j = 1, 2, 3, 4.$$

The prior on the state vector is assumed to be diffuse: $\hat{x}_{0|0} = 0$ and $P_{0|0}^{-1} = 0$.

If we want to implement the fully Bayesian analysis of the basic structural model by the data augmentation algorithm, we have to choose a starting approximation $g_0(\theta)$, e.g.

$$g_0(\theta) = p_{\text{IG}}(\theta_1; 3, 0.002) p_{\text{IG}}(\theta_2; 3, 0.0002) p_{\text{IG}}(\theta_3; 3, 0.0002) p_{\text{IG}}(\theta_4; 3, 0.02).$$

The expectation of this starting approximation $\hat{\theta}_0 = (0.001, 0.0001, 0.0001, 0.01)$ is comparable with the starting value necessary for maximum likelihood estimation.

During the first 50 iterations the number M_n of sampled paths is equal to 100. Figure 2 monitors convergence of the algorithm by plotting 95%-highest probability density (95%-HPD) regions of the approximate marginal posteriors $g_n(\theta_1), \dots, g_n(\theta_4)$ over the number of iterations. After 50 iterations the steady state seems to have been reached and the final 10 iterations are carried out with $M_n = 2000$. The monitor nicely demonstrates how this increase reduces the variances of the density estimate.

The posterior $p(\theta | y^{92})$ of the unknown variances finally is approximated by $g_{60}(\theta)$. Figure 3 shows the bivariate marginals of this approximate posterior together with the contours. It is interesting to observe that the shape of the posterior is quite different for the different variances. $p(\theta_4 | \theta_j, y^{92})$ is rather symmetric and clearly shifted away from 0. $p(\theta_2 | \theta_j, y^{92})$ is rather skew and slightly shifted away from 0. $p(\theta_1 | \theta_j, y^{92})$ and $p(\theta_3 | \theta_j, y^{92})$ are extremely skew and concentrated near 0.

Comparable statements may be derived from the 95%-highest probability density sample (95%-HPS) of the unknown variances shown in Figure 4. The $(1 - \alpha)$ -HPS is a sampling-based approximation to the $(1 - \alpha)$ -HPD region. To obtain a $(1 - \alpha)$ -HPS we draw a sample $\theta^{(m)}$ of size M_H with $M_H \alpha \in \mathbb{N}$ from the approximate posterior $p(\theta | y^N)$ and use a multi-process filter (Harrison and Stevens, 1976) to compute the functional values p_m of the

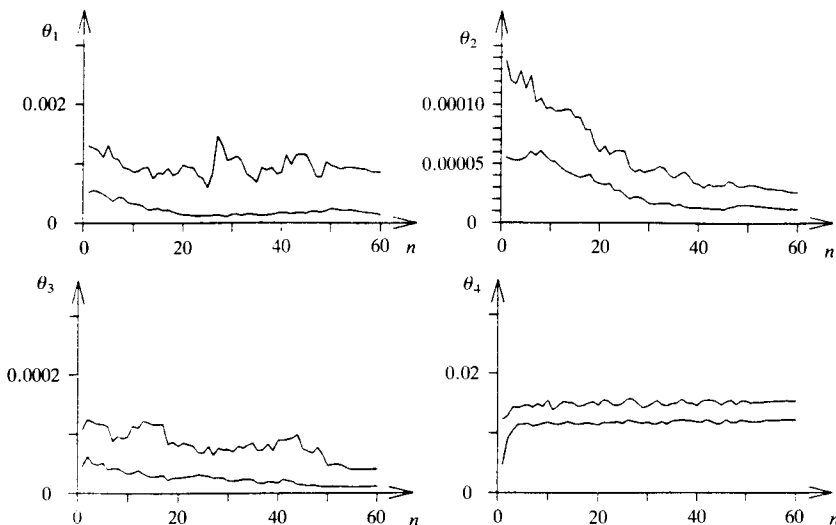


FIGURE 2. Monitoring convergence of the approximate posterior.

exact non-normalized posterior density. We then define the $(1 - \alpha)$ -HPS by

$$\Theta_\alpha = \{\theta^{(m)}: p_m > p_{(\alpha M_H)}\}$$

where $p_{(1)}, \dots, p_{(M_H)}$ denotes the ordered set of $p_1, \dots, p_{M_H}$. For such a set the following properties are obvious:

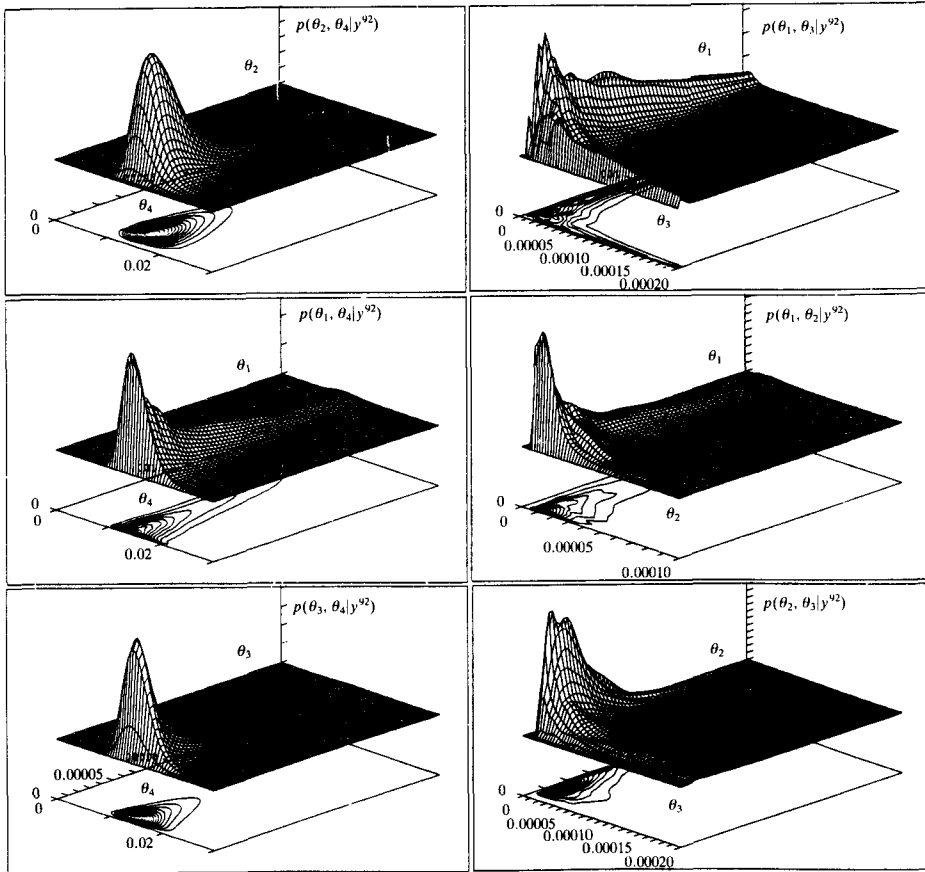
$$h(\theta \in \Theta_\alpha | y^N) = 1 - \alpha$$

$$p(\theta^{(m)} | y^N) \geq p(\theta^{(m')} | y^N) \quad \forall \theta^{(m)} \in \Theta_\alpha, \theta^{(m')} \notin \Theta_\alpha$$

where $h(\cdot)$ denotes the relative frequency of an event. To visualize the 95%-HPS for the present case study it has been projected to the six different (θ_i, θ_j) planes (Figure 4). These figures, too, indicate that θ_2 and θ_4 are likely to take positive values, whereas θ_1 and θ_3 are concentrated close to 0.

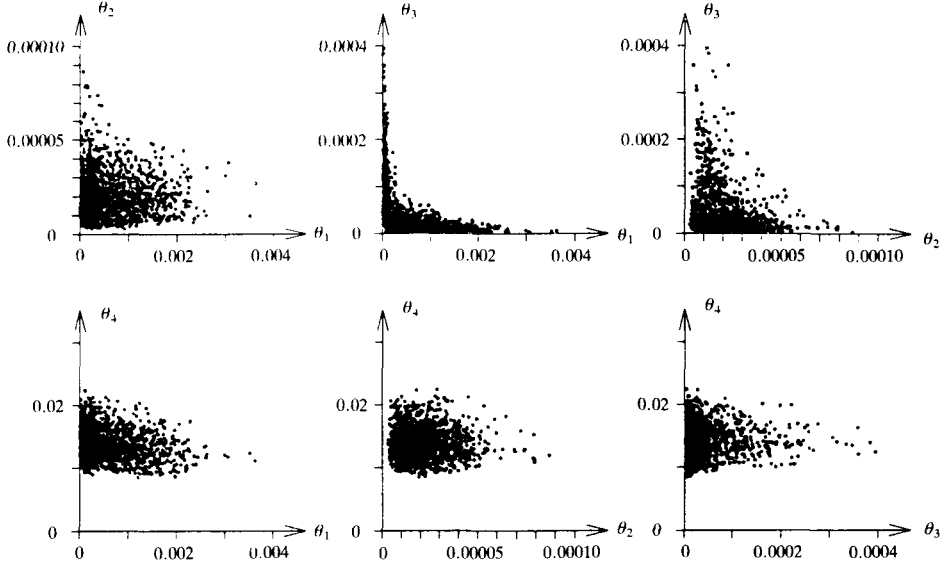
To summarize, the fully Bayesian analysis leads to the conclusion that any description of the uncertainty about the parameter which is derived from asymptotic normality is inadequate for this data set. As far as θ_2 and θ_4 are concerned this failure seems to be caused by the lack of a sufficient number of observations. For θ_1 and θ_3 there is some support for the hypothesis that either one or both of them are 0 and asymptotic normality would not hold even for a higher number of observations.

More formal testing of this hypothesis against the hypothesis that all variances are positive is non-standard. It is carried out through Bayesian model discrimination in Frühwirth-Schnatter (1994) and leads for the UK coal consumption series in fact to a decision in favour of the hypothesis $\theta_1 = \theta_3 = 0$.

FIGURE 3. Bivariate marginals of the posterior p.d.f. $p(\theta|y^{92})$.

6. CONCLUDING REMARKS

In this paper a data augmentation algorithm has been suggested for approximating posterior p.d.f.s for dynamic linear models with unknown hyperparameter. We confined ourselves to a particular subclass of such models called the d -inverse-gamma class. The relation between data augmentation methods and dynamic models certainly holds in a more general class. The algorithm suggested for practical implementation may be applied to *any* dynamic linear model with unknown hyperparameter provided the conditional density $p(\theta|x^N, y^N)$ is of such a form that we are able to sample from it. The restriction to the d -inverse-gamma class is mainly due to the fact that so far we have proved the regularity conditions sufficient for convergence of the algorithm for this class only. However, we are quite confident that the algorithm converges for a wider class of dynamic linear models.

FIGURE 4. Projection of a 95%-HPS of $\theta|y^{92}$.

APPENDIX: PROOFS

PROOF OF LEMMA 1. Let us prove the lemma for the case (9). First we assume that the last m components of $\tilde{\theta}$ are reduced to 0: $\theta_j = \tilde{\theta}_j$, $1 \leq j \leq r - m$, $\theta_{r-m+1} = \theta_{d-m} = \tilde{\theta}_d$. x^N is realizable iff the last m components of $w_t = B_t^{-1}(y^{t-1})(x_t - F_t x_{t-1})$ are equal to zero. For such a realizable path the following holds:

$$\prod_{t=1}^N p(x_t | x_{t-1}, \theta) \propto \prod_{j=1}^{r-m} \left(\frac{1}{\theta_j} \right)^{1/2} \exp \left[-\frac{1}{2} \sum_{t=1}^N [B_t^{-1}(y^{t-1})(x_t - F_t x_{t-1})]_j^2 \frac{1}{\theta_j} \right].$$

From this result and (3) we see that $p(\theta | x^N, y^N)$ is the product of $(d - m)$ -inverse-gamma densities. Thus (13) is obvious now from (12).

If $\tilde{\theta}$ does not split directly into $(\theta, 0)$ we choose a suitable permutation matrix Π to obtain a new parametrization $\tilde{\theta}' = \Pi \tilde{\theta} = (\theta, 0)$ of the hyperparameter. Then the same proof as above is applied to the new parametrization. The case (8) is proved in a similar manner.

PROOF OF LEMMA 2. For each $j = 1, \dots, d$ the p.d.f. $p(\theta_j | x^N, y^N)$ is bounded by the functional value of this density at the mode. This functional value has an upper bound which is independent of x^N :

$$p(\theta_j | x^N, y^N) \leq \frac{c(\alpha_N^{(j)})}{\beta_0^{(j)} + b_N^{(j)}(x^N, y^N)} \leq \frac{c(\alpha_N^{(j)})}{\beta_0^{(j)}} \quad c(\alpha) = \frac{(\alpha + 1)^{\alpha+1}}{\Gamma(\alpha)} \exp[-(\alpha + 1)]. \quad (25)$$

This bound is finite because of Condition (C4) and $c(\alpha) < \infty$ for $\alpha < \infty$. Thus Condition (C1) follows from

$$K(\theta, \theta') = \int p(\theta | x^N, y^N) p(x^N | y^N, \theta') dx^N \leq \prod_{j=1}^d \frac{c(\alpha_N^{(j)})}{\beta_0^{(j)}} < \infty \quad \forall \theta, \theta' \in \Theta.$$

As each partial derivative of the conditional posterior $p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N)$ of the hyperparameter is continuous in $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ the conditional posterior $p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N)$ of the hyperparameter is differentiable $\forall \boldsymbol{\theta} \in \boldsymbol{\Theta}$. Now take $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ arbitrarily. Let $M > 0$ be some constant. Then for any $\tilde{\boldsymbol{\theta}}$ with $\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}\| \leq M$ the following representation holds with $\xi \in \tilde{\boldsymbol{\theta}}\boldsymbol{\Theta}$:

$$p(\tilde{\boldsymbol{\theta}}|\mathbf{x}^N, \mathbf{y}^N) - p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N) = \sum_{j=1}^d \frac{\partial p(\xi|\mathbf{x}^N, \mathbf{y}^N)}{\partial \theta_j} (\tilde{\theta}_j - \theta_j). \quad (26)$$

The first derivative of an inverted gamma density is bounded:

$$\begin{aligned} \frac{\partial p(\theta_j|\mathbf{x}^N, \mathbf{y}^N)}{\partial \theta_j} &\leq \frac{c'(\alpha_N^{(j)})}{\{\beta_0^{(j)} + b_N^{(j)}(\mathbf{x}^N, \mathbf{y}^N)\}^2} \leq \frac{c'(\alpha_N^{(j)})}{(\beta_0^{(j)})^2} < \infty \\ c'(\alpha) &= \frac{(\alpha + 2)^{(\alpha+2)/2}}{\Gamma(\alpha)} \left\{ \frac{\alpha + 1}{(\alpha + 2)^{1/2} - 1} \right\}^{\alpha+3} \exp \left\{ -\frac{(\alpha + 1)(\alpha + 2)^{1/2}}{(\alpha + 2)^{1/2} + 1} \right\}. \end{aligned} \quad (27)$$

This bound is finite because of Condition (C4) and $c'(\alpha) < \infty$ for $\alpha < \infty$. Combining (25) and (27) we obtain that the first derivatives in formula (26) are bounded:

$$\frac{\partial p(\xi|\mathbf{x}^N, \mathbf{y}^N)}{\partial \theta_j} \leq \left\{ \prod_{i=1, i \neq j}^d \frac{c(\alpha_N^{(i)})}{\beta_0^{(i)}} \right\} \frac{c'(\alpha_N^{(j)})}{(\beta_0^{(j)})^2} =: c_j''(N, \boldsymbol{\alpha}_0, \boldsymbol{\beta}_0).$$

The bound $c_j''(N, \boldsymbol{\alpha}_0, \boldsymbol{\beta}_0)$ depends only on the number N of observations and on the parameters $\boldsymbol{\alpha}_0 = (\alpha_0^{(1)}, \dots, \alpha_0^{(d)})$ and $\boldsymbol{\beta}_0 = (\beta_0^{(1)}, \dots, \beta_0^{(d)})$ of the prior of the hyperparameter. Therefore,

$$\begin{aligned} |K(\boldsymbol{\theta}, \boldsymbol{\theta}') - K(\tilde{\boldsymbol{\theta}}, \boldsymbol{\theta}')| &\leq \int |p(\tilde{\boldsymbol{\theta}}|\mathbf{x}^N, \mathbf{y}^N) - p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N)| p(\mathbf{x}^N|\boldsymbol{\theta}', \mathbf{y}^N) d\mathbf{x}^N \\ &\leq M \sum_{j=1}^d c_j''(N, \boldsymbol{\alpha}_0, \boldsymbol{\beta}_0). \end{aligned}$$

Thus $\forall \boldsymbol{\theta} \in \boldsymbol{\Theta}$ and $\forall \varepsilon > 0$ the choice of

$$\delta_\varepsilon = \frac{\varepsilon}{\sum_{j=1}^d c_j''(N, \boldsymbol{\alpha}_0, \boldsymbol{\beta}_0)}$$

leads to $\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}\| \leq \delta_\varepsilon \Rightarrow |K(\boldsymbol{\theta}, \boldsymbol{\theta}') - K(\tilde{\boldsymbol{\theta}}, \boldsymbol{\theta}')| \leq \varepsilon \quad \forall \boldsymbol{\theta}' \in \boldsymbol{\Theta}$. This proves Condition (C2).

The p.d.f.s $p(\theta_j|\mathbf{x}^N, \mathbf{y}^N)$ are strictly positive on $\mathbb{R}_+$. Thus $p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N)$ is strictly positive on $\boldsymbol{\Theta}$. Now take $\boldsymbol{\theta}_0 \in \boldsymbol{\Theta}$. As $\boldsymbol{\Theta}$ is open $\forall \varepsilon > 0 \exists$ open neighbourhood $U_\varepsilon(\boldsymbol{\theta}_0) \subset \boldsymbol{\Theta}$. As $p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N)$ is strictly positive $\forall \boldsymbol{\theta} \in U_\varepsilon(\boldsymbol{\theta}_0)$, we define $c := \inf_{\boldsymbol{\theta} \in U_\varepsilon(\boldsymbol{\theta}_0)} p(\boldsymbol{\theta}|\mathbf{x}^N, \mathbf{y}^N) > 0$. Then $K(\boldsymbol{\theta}, \boldsymbol{\theta}')$ is positive on $U_\varepsilon(\boldsymbol{\theta}_0)$. Thus Condition (C3) holds.

PROOF OF PROPOSITION 1. For a fully dynamic model with regular $\mathbf{Q}_t(\boldsymbol{\theta})$, minus twice the log of the right-hand side of Equation (20) is proportional to

$$(\mathbf{x}_{t+1} - \mathbf{F}_{t+1}\mathbf{x}_t)^\top \mathbf{Q}_{t+1}(\boldsymbol{\theta})^{-1} (\mathbf{x}_{t+1} - \mathbf{F}_{t+1}\mathbf{x}_t) + \{\mathbf{x}_t - \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})\}^\top \mathbf{P}_{t|t}(\boldsymbol{\theta})^{-1} \{\mathbf{x}_t - \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})\}.$$

Completing the squares gives the desired result.

PROOF OF PROPOSITION 2. From (21) it follows that

$$\Pi \mathbf{x}_{t+1} = \Pi(\mathbf{F}_{t+1}\mathbf{x}_t + \mathbf{w}_{t+1}) = \mathbf{z}_t + \begin{pmatrix} \mathbf{w}_{t+1}^* \\ \mathbf{0} \end{pmatrix} \quad \mathbf{w}_{t+1}^* \sim \mathbf{N}(\mathbf{0}, \mathbf{Q}_{t+1}^*(\boldsymbol{\theta})).$$

Therefore the last $r - d$ components of $\mathbf{z}_t$ are deterministic and equal to $\Pi^0 \mathbf{x}_{t+1}$ given

$\mathbf{x}_{t+1}$. Let $\mathbf{z}_t^*$ denote the remaining stochastic components. From Bayes' theorem we obtain that minus twice the log of $p(\mathbf{z}_t^*|\mathbf{x}_{t+1}, y')$ is proportional to

$$(\Pi^* \mathbf{x}_{t+1} - \mathbf{z}_t^*)^T \mathbf{Q}_{t+1}^*(\boldsymbol{\theta})^{-1} (\Pi^* \mathbf{x}_{t+1} - \mathbf{z}_t^*) \\ + \{\mathbf{z}_t^* - \Pi^* \mathbf{F}_{t+1} \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})\}^T \{\Pi^* \mathbf{F}_{t+1} \mathbf{P}_{t|t}(\boldsymbol{\theta}) (\Pi^* \mathbf{F}_{t+1})^T\}^{-1} \{\mathbf{z}_t^* - \Pi^* \mathbf{F}_{t+1} \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})\}.$$

Completing the squares gives the desired result.

PROOF OF PROPOSITION 3. Minus twice the log of the right-hand side of Equation (20) is proportional to

$$(\mathbf{x}_t - \mathbf{F}_{t+1}^{-1} \mathbf{x}_{t+1})^T \mathbf{W}_{t+1}(\boldsymbol{\theta})^{-1} (\mathbf{x}_t - \mathbf{F}_{t+1}^{-1} \mathbf{x}_{t+1}) + \{\mathbf{x}_t - \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})\}^T \mathbf{P}_{t|t}(\boldsymbol{\theta})^{-1} \{\mathbf{x}_t - \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})\}$$

where $\mathbf{W}_{t+1}(\boldsymbol{\theta})$ is the same as in (22). Completing the squares gives

$$-2 \log \{p(\mathbf{x}_t|\mathbf{x}_{t+1}, y', \boldsymbol{\theta})\} \propto \{\mathbf{x}_t - \hat{\mathbf{x}}_{t|N}(\mathbf{x}_{t+1}, \boldsymbol{\theta})\}^T \mathbf{P}_{t|N}(\boldsymbol{\theta})^{-1} \{\mathbf{x}_t - \hat{\mathbf{x}}_{t|N}(\mathbf{x}_{t+1}, \boldsymbol{\theta})\}$$

with

$$\mathbf{P}_{t|N}(\boldsymbol{\theta}) = \{\mathbf{W}_{t+1}(\boldsymbol{\theta})^{-1} + \mathbf{P}_{t|t}(\boldsymbol{\theta})^{-1}\}^{-1} \\ = \mathbf{W}_{t+1}(\boldsymbol{\theta}) \{\mathbf{I} + \mathbf{C}_{t|t}(\boldsymbol{\theta}) \mathbf{W}_{t+1}(\boldsymbol{\theta})\}^{-1} \\ = \mathbf{W}_{t+1}(\boldsymbol{\theta}) \mathbf{V}_{t+1}(\boldsymbol{\theta})$$

and

$$\hat{\mathbf{x}}_{t|N}(\mathbf{x}_{t+1}, \boldsymbol{\theta}) = \mathbf{P}_{t|N}(\boldsymbol{\theta}) \{\mathbf{W}_{t+1}(\boldsymbol{\theta})^{-1} \mathbf{F}_{t+1}^{-1} \mathbf{x}_{t+1} + \mathbf{P}_{t|t}(\boldsymbol{\theta})^{-1} \hat{\mathbf{x}}_{t|t}(\boldsymbol{\theta})\} \\ = \mathbf{V}_{t+1}(\boldsymbol{\theta})^T \mathbf{F}_{t+1}^{-1} \mathbf{x}_{t+1} + \mathbf{P}_{t|N}(\boldsymbol{\theta}) \mathbf{m}_{t|t}(\boldsymbol{\theta})$$

as

$$\mathbf{V}_{t+1}(\boldsymbol{\theta}) \mathbf{W}_{t+1}(\boldsymbol{\theta})^{-1} = \mathbf{W}_{t+1}(\boldsymbol{\theta})^{-1} \mathbf{V}_{t+1}(\boldsymbol{\theta})^T.$$

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